

OS Lab 7 Task

1)

'01a-input.txt'

```

arch-host
CPU-Scheduling-Algorithms main ? X cat testcases/0
01a-input.txt 02b-input.txt 04a-input.txt 05b-input.txt 07a-input.txt 08b-input.txt
01a-output.txt 02b-output.txt 04a-output.txt 05b-output.txt 07a-output.txt 08b-output.txt
01b-input.txt 03a-input.txt 04b-input.txt 06a-input.txt 07b-input.txt 09a-input.txt
01b-output.txt 03a-output.txt 04b-output.txt 06a-output.txt 07b-output.txt 09a-output.txt
02a-input.txt 03b-input.txt 05a-input.txt 06b-input.txt 08a-input.txt 09b-input.txt
02a-output.txt 03b-output.txt 05a-output.txt 06b-output.txt 08a-output.txt 09b-output.txt

CPU-Scheduling-Algorithms main ? X cat testcases/01
01a-input.txt 01a-output.txt 01b-input.txt 01b-output.txt

CPU-Scheduling-Algorithms main ? X cat testcases/01a-
...input.txt ...output.txt

CPU-Scheduling-Algorithms main ? X cat testcases/01a-input.txt
trace
1
20
5
A,0,3
B,2,6
C,4,4
D,6,5
E,8,2

CPU-Scheduling-Algorithms main ? > ./lab4
trace
1
20
5
A,0,3
B,2,6
C,4,4
D,6,5
E,8,2
FCFS 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0
-----
A |*|*|*| | | | | | | | | | | | | | | | |
B | | |*|*|*|*|*|*| | | | | | | | | | |
C | | | | |*|*|*|*|*|*| | | | | | | | |
D | | | | | | | | | | | | |*|*|*|*|*|*|
E | | | | | | | | | | | | | | | | |*|*|
-----

CPU-Scheduling-Algorithms main ? >

```

2)

'01b-input.txt'

```

CPU-Scheduling-Algorithms main ? > cat testcases/01b-
...input.txt    ...output.txt

CPU-Scheduling-Algorithms main ? > cat testcases/01b-input.txt
stats
1
20
5
A,0,3
B,2,6
C,4,4
D,6,5
E,8,2

CPU-Scheduling-Algorithms main ? >
Display all 3613 possibilities? (y or n)

CPU-Scheduling-Algorithms main ? > ./lab4
stats
1
20
5
A,0,3
B,2,6
C,4,4
D,6,5
E,8,2
FCFS
Process      | A | B | C | D | E |
Arrival      | 0 | 2 | 4 | 6 | 8 |
Service      | 3 | 6 | 4 | 5 | 2 | Mean|
Finish       | 3 | 9 | 13| 18| 20|-----|
Turnaround   | 3 | 7 | 9 | 12| 12| 8.60|
NormTurn     | 1.00| 1.17| 2.25| 2.40| 6.00| 2.56|

CPU-Scheduling-Algorithms main ? > █

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3)

```

CPU-Scheduling-Algorithms main ? X cat testcases/02a-input.txt
trace
2-1
20
5
A,0,3
B,2,6
C,4,4
D,6,5
E,8,2

```

```

CPU-Scheduling-Algorithms main ? > ./lab4

```

```

trace
2-1
20
5
A,0,3
B,2,6
C,4,4
D,6,5
E,8,2
RR-1  0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0
-----
A      |*|*|. |*| | | | | | | | | | | | | | | |
B      | | |*|. |*|. |*|. |. |. |. |*|. |. |. |*| | |
C      | | | | |. |*|. |. |*|. |. |. |. |*|. |. |. |*| |
D      | | | | | | |. |*|. |. |. |*|. |. |. |*|. |. |*|*|
E      | | | | | | | | |. |. |*|. |. |. |*| | | | | |
-----

```

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```

Round Robin with Quantum=1

Process	A	B	C	D	E	
Arrival	0	2	4	6	8	
Service	3	6	4	5	2	Mean
Finish	4	18	17	20	15	-----
Turnaround	4	16	13	14	7	10.80
NormTurn	1.33	2.67	3.25	2.80	3.50	2.71

```

CPU-Scheduling-Algorithms main ? > cat testcases/03a-input.txt
trace
2-4
20
5
A,0,3
B,2,6
C,4,4
D,6,5
E,8,2

```

```

CPU-Scheduling-Algorithms main ? > ./lab4

```

```

trace
2-4
20
5
A,0,3
B,2,6
C,4,4
D,6,5
E,8,2
RR-4  0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0
-----
A      |*|*|*| | | | | | | | | | | | | | | | |
B      | | | |*|*|*|*|.|.|.|.|.|.|.|.|.|.|.|.|.|.|.
C      | | | | | |*|*|*|*|.|.|.|.|.|.|.|.|.|.|.|.|.
D      | | | | | | | |*|*|*|*|.|.|.|.|.|.|.|.|.|.
E      | | | | | | | | | |*|*|*|*|.|.|.|.|.|.|.|.
-----

```

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CPU-Scheduling-Algorithms main ? > █

```

Round Robin with Quantum=4

Process	A	B	C	D	E
Arrival	0	2	4	6	8
Service	3	6	4	5	2
Mean					
Finish	3	17	11	20	19
Turnaround	3	15	7	14	11
10.00					
NormTurn	1.00	2.50	1.75	2.80	5.50
2.71					

```
CPU-Scheduling-Algorithms main ? > cat testcases/04a-input.txt
```

```
trace
```

```
3
```

```
20
```

```
5
```

```
A,0,3
```

```
B,2,6
```

```
C,4,4
```

```
D,6,5
```

```
E,8,2
```

```
CPU-Scheduling-Algorithms main ? > ./lab4
```

```
trace
```

```
3
```

```
20
```

```
5
```

```
A,0,3
```

```
B,2,6
```

```
C,4,4
```

```
D,6,5
```

```
E,8,2
```

```
SPN  0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0
-----
A      |*|*|*| | | | | | | | | | | | | | | | |
B      | | | |*|*|*|*|*|*| | | | | | | | | | |
C      | | | | | | | | | | | |*|*|*|*| | | | |
D      | | | | | | | | | | | | | | |*|*|*|*|*|
E      | | | | | | | | |*|*| | | | | | | | | |
-----
```

```
CPU-Scheduling-Algorithms main ? > █
```

Shortest Process Next

Process	A	B	C	D	E	
Arrival	0	2	4	6	8	
Service	3	6	4	5	2	Mean
Finish	3	9	15	20	11	-----
Turnaround	3	7	11	14	3	7.60
NormTurn	1.00	1.17	2.75	2.80	1.50	1.84

```

CPU-Scheduling-Algorithms main ? > cat testcases/05a-input.txt
trace
4
20
5
A,0,3
B,2,6
C,4,4
D,6,5
E,8,2

CPU-Scheduling-Algorithms main ? > ./lab4
trace
4
20
5
A,0,3
B,2,6
C,4,4
D,6,5
E,8,2
SRT  0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0
-----
A    |*|*|*| | | | | | | | | | | | | | | | | |
B    | | | |*| | | | | | | |*|*|*|*|*| | | | |
C    | | | | |*|*|*|*| | | | | | | | | | | | |
D    | | | | | | | | | | | | | | | |*|*|*|*|*|
E    | | | | | | | | |*|*| | | | | | | | | | |
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CPU-Scheduling-Algorithms main ? > █

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Shortest Remaining Time First

Process	A	B	C	D	E	
Arrival	0	2	4	6	8	
Service	3	6	4	5	2	Mean
Finish	3	15	8	20	10	----
Turnaround	3	13	4	14	2	7.20
NormTurn	1.00	2.17	1.00	2.80	1.00	1.59

```

CPU-Scheduling-Algorithms main ? > cat testcases/06a-input.txt
trace
5
20
5
A,0,3
B,2,6
C,4,4
D,6,5
E,8,2

CPU-Scheduling-Algorithms main ? > ./
lab4*      testcases/

CPU-Scheduling-Algorithms main ? > ./lab4
trace
5
20
5
A,0,3
B,2,6
C,4,4
D,6,5
E,8,2
HRRN      0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0
-----
A          |*|*|*| | | | | | | | | | | | | | | | |
B          | | | |*|*|*|*|*| | | | | | | | | | |
C          | | | | | | | | | |*|*|*|*| | | | | |
D          | | | | | | | | | | | | | | | |*|*|*|*|
E          | | | | | | | | | | | | | |*|*| | | | |
-----

CPU-Scheduling-Algorithms main ? > █

```

Highest Response Ratio Next

Process	A	B	C	D	E	
Arrival	0	2	4	6	8	
Service	3	6	4	5	2	Mean
Finish	3	9	13	20	15	-----
Turnaround	3	7	9	14	7	8.00
NormTurn	1.00	1.17	2.25	2.80	3.50	2.14

```

CPU-Scheduling-Algorithms main ? > cat testcases/07a-input.txt
trace
6
20
5
A,0,3
B,2,6
C,4,4
D,6,5
E,8,2

CPU-Scheduling-Algorithms main ? > ./lab4
trace
6
20
5
A,0,3
B,2,6
C,4,4
D,6,5
E,8,2
FB-1  0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0
-----
A      |*|*|. |*| | | | | | | | | | | | | | | | |
B      | | | |*|. |*|. |. |. |. |. |*|. |. |*|. |. |*|. |
C      | | | | |*|. |*|. |. |. |. |. |*|. |. |*|. | | |
D      | | | | | |*|. |*|. |. |. |. |. |*|. |. |*|. |
E      | | | | | | | |*|. |*| | | | | | | | | | |
-----

CPU-Scheduling-Algorithms main ? > █

```

Feedback where all queues have Quantum=1

Process	A	B	C	D	E	
Arrival	0	2	4	6	8	
Service	3	6	4	5	2	Mean
Finish	4	20	16	19	11	-----
Turnaround	4	18	12	13	3	10.00
NormTurn	1.33	3.00	3.00	2.60	1.50	2.29

```

CPU-Scheduling-Algorithms main ? > cat testcases/08a-input.txt
trace
7
20
5
A,0,3
B,2,6
C,4,4
D,6,5
E,8,2

```

```

CPU-Scheduling-Algorithms main ? > ./lab4

```

```

trace
7
20
5
A,0,3
B,2,6
C,4,4
D,6,5
E,8,2
FB-2i 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0
-----
A      |*|*|. |*| | | | | | | | | | | | | | | | |
B      | | |*|. |*|*|. |. |. |. |. |. |. |*|*|*| | | |
C      | | | | |*|. |. |. |. |*|*|. |. |. |. |. |*| | |
D      | | | | | | |*|. |. |. |*|*|. |. |. |. |*|*| |
E      | | | | | | | | |*|. |. |. |. |*| | | | | | |
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```

Feedback where Quantum=2i

Process	A	B	C	D	E	
Arrival	0	2	4	6	8	
Service	3	6	4	5	2	Mean
Finish	4	17	18	20	14	----
Turnaround	4	15	14	14	6	10.60
NormTurn	1.33	2.50	3.50	2.80	3.00	2.63

```
CPU-Scheduling-Algorithms main ? > ./lab4
trace
2-2
20
5
A,0,3
B,2,6
C,4,4
D,6,5
E,8,2
```

CPU-Scheduling-Algorithms main ? >

Process	A	B	C	D	E	
Arrival	0	2	4	6	8	
Service	3	6	4	5	2	Mean
Finish	5	17	13	20	15	-----
Turnaround	5	15	9	14	7	10.00
NormTurn	1.67	2.50	2.25	2.80	3.50	2.54